Indian Association for the Cultivation of Science

(Deemed to be University under *de novo* Category)

Master's/Integrated Master's-PhD Program/Integrated Bachelor's-Master's Program/PhD Course

End-Semester Examination-Autumn 2025

Subject: Discrete Mathematics — *Subject Code(s): COM-2101/COM-4102*

Full Marks: 50 — *Time Allotted: 3 h*

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1. Prove that T is a tree if and only if between any two vertices of G, there is exactly one path. 3

2. Construct a graph with 8 vertices which has at least one bridge and at least two distinct spanning trees. 3

3. Justify the statement – A connected graph is semi-Eulerian if and only if it has exactly two vertices of odd degree. Construct a semi-Eulerian graph with 8 vertices and demonstrate Fleury's algorithm to find an Eulerian Path. 5

4. How many Hamiltonian Paths are present in K_3, K_4, and K_5. Can you generalize it for K_n. Note that, a Hamiltonian path can start from any vertex. 5

5. Let G_1 and G_2 be two graphs defined as follows:
G_1 has vertices { a,b,c,d} with edges { (a,b),(b,c),(c,d),(d,a),(a,c)} .G_2 has vertices {1,2,3,4\} with edges {(1,2),(2,3),(3,4),(4,1),(2,4)}. .Are G_1 and G_2 isomorphic? Give reasons in support of your answer. Which one do you think is an easy computational problem to solve – a) checking whether two graphs are isomorphic or not, b) checking whether two trees are isomorphic or not. 5

6. An undirected graph has 5 vertices and sum total of all the degrees of the vertices of the graph is 8. How many connected graphs can you construct. Make an optimized amendment by increasing the total degree of all the vertices to make a connected graph which is both Eulerian and Hamiltonian. 5

7. Define MacMahon's partition functions. Compute MacMahon's partition functions $M_1(n)$ and $M_2(n)$ for the integers 5 and 6 and use these functions to evaluate $(n^2 - 3n + 2).M_1(n) - 8.M_2(n)$. What is the significance that we can draw from the values of this function. $2 + 4 + 2 = 8$

8. How many positive integers less than 500 can be expressed as product of three distinct primes. 5

9. A company has **4 workers** W_1,W_2,W_3,W_4 and **4 tasks** T_1,T_2,T_3,T_4. Each worker can perform each task, but the cost (in hours) differs. The cost matrix is given below:

	T_1	**T_2**	**T_3**	**T_4**
W_1	9	2	7	8
W_2	6	4	3	7
W_3	5	8	1	8
W_4	7	6	9	4

Formulate this as an **assignment problem** and determine the optimal assignment of workers to tasks that minimizes the total cost. How do you think, to approach this problem if there are N tasks and M workers (N > M). Assume in this case that a worked can be assigned to a Task which is not started yet, only when a Worker has finished his previously assigned task. $3 + 3 = 6$

10. Let N and M be two positive integers with number of divisors equal to P and Q respectively. What do you think the number of divisors of N.M would be. 5